In-DMU: Modeling Uncertainty in Interferometric SAR-Based Deformation Monitoring

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***Abstract*—Quantifying the precision of interferometric synthetic aperture radar (InSAR) deformation monitoring is a fundamental aspect of ensuring its reliability in operational applications. Traditional uncertainty estimation methods, which are based on phase noise and error propagation laws, tend to overestimate monitoring precision. In this study, we systematically investigate the combined effects of coherence, deformation magnitude, SAR wavelength, and spatial resolution on monitoring uncertainty through multiparameter-controlled experiments. Based on this, a wavelength–coherence integrated function model, referred to as the InSAR deformation monitoring uncertainty (In-DMU) model, was developed to estimate the precision of the deformation measurements. To rigorously define the applicability of the model, a Pettitt test based on sliding-window standard deviation (Pt-SWStd) is used to detect change points in the error distribution, thus establishing critical gradient thresholds of 6.433 mm/m (L band), 1.407 mm/m (C band), and 0.456 mm/m (X band). Beyond these thresholds, phase-unwrapping constraints lead to the retrieval breakdown regime, limiting reliable deformation estimation. Furthermore, the influence of multilooking processing on the In-DMU model was systematically quantified. To assess practical applicability, the In-DMU model was validated across diverse observational scenarios using data from five representative SAR satellites: ALOS-1 (L band), ALOS-2 (L band), Radarsat-2 (C band), Sentinel-1 (C band), and TerraSAR-X (X band). The In-DMU model provides a universal tool for a priori estimation of InSAR monitoring precision, offering valuable guidance for SAR data selection and optimization of multilooking configurations.**

***Index Terms*—Coherence, deformation, interferometric synthetic aperture radar (InSAR), multilooking, uncertainty estimate.**

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I. Introduction

SURFACE deformation monitoring plays a crucial role in deciphering solid Earth dynamics, elucidating deformation mechanisms, and enabling early warnings of geological hazards [1], [2], [3]. It serves as a key foundation to ensure environmental protection and support sustainable development [4]. Among various geodetic techniques, interferometric synthetic aperture radar (InSAR) has become indispensable in modern geodesy due to its wide-area coverage, high spatiotemporal resolution, and vast archive of historical data [5]. Currently, InSAR has been widely applied in 3-D topographic reconstruction [6], land subsidence monitoring [7], mining-induced subsidence assessment [8], [9], earthquake rupture inversion [10], landslide identification [11], and monitoring volcanic activity [12]. It effectively overcomes the inherent limitations of traditional geodetic techniques, such as restricted spatial coverage and low operational efficiency [13], [14]. Recent advancements in the SAR platform performance, combined with the development of multiband, multipolarization, and multimode imaging systems, have led to significant breakthroughs in SAR satellites [15], [16]. These innovations collectively enable high-resolution wide-swath imaging, synchronized multiplatform constellation operations, and high-frequency revisit observations [17]. As a result, InSAR's capability to characterize multimechanism deformation processes has been significantly improved, thereby enabling its broader application in pivotal areas, such as global change research and disaster mitigation engineering [18].

Determining the precision of InSAR deformation measurements is a critical prerequisite for fully utilizing the

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potential applications of the InSAR technology [19]. Currently, methods for assessing the precision of InSAR-derived deformation measurements are classified into internal evaluation and external validation [20]. Internal evaluation, used predominantly in time-series InSAR analyses, quantifies precision using phase residual standard errors or displacement velocity standard deviations (STDs) [21], [22], [23]. The effectiveness of this approach is largely determined by the quality of the monitoring pixels [24]. However, its reliability diminishes in scenarios involving nonlinear deformation. In contrast, the external validation establishes monitoring precision by cross-verifying InSAR-derived displacements against ground-based measurements (e.g., GNSS and leveling) [25], [26]. This approach, however, imposes stringent constraints on reference data sources, particularly in terms of spatial sampling density, temporal synchronization, and data precision [27]. These factors frequently restrict its operational feasibility in large-scale deployments.

The above methods present both strengths and weaknesses. A critical limitation is their inability to estimate the InSAR monitoring uncertainty in the target regions prior to SAR data processing. The lack of a priori precision evaluation introduces substantial uncertainties during SAR data selection [28]. This deficiency not only undermines the reliability of derived deformation products and subsequent geophysical interpretations but also results in unnecessary consumption of SAR data resources and excessive computational overhead [29]. Recent advancements in multiplatform SAR constellations enable the strategic selection of SAR datasets with optimal wavelengths, incidence angles, and spatial resolutions to suit specific monitoring scenarios [30], [31]. Therefore, developing a priori model to estimate the uncertainty of InSAR-derived deformation is essential for selecting SAR data based on specific requirements.

Coherence, a pivotal parameter that influences the precision of InSAR deformation monitoring, fully reflects the effects of scatterer temporal variations (i.e., temporal decorrelation) and interferometric baseline geometry (i.e., spatial decorrelation) [32], [33]. The unbiased coherence estimation remains a challenging task in practical applications due to the heterogeneity of SAR pixels within the estimation window and the inherent bias of coherence estimators [34], [35], [36]. Nevertheless, biased coherence estimates can still be used effectively in various applications, such as the selection of high-quality phase points and the optimization of interferometric phases [37], [38], [39], [40]. In addition, the nonlinear functional relationship between the coherence level and the statistical distribution of the interferometric phases was demonstrated, which allowed the estimation of wrapped-phase uncertainty based on coherence [38], [41]. However, the nonlinear transformation introduced during phase unwrapping prevents a direct correspondence between the wrapped-phase uncertainty and the unwrapped-phase uncertainty [21]. This nonlinear propagation effect introduces systematic errors when estimating deformation monitoring uncertainties directly from wrapped-phase uncertainties using error propagation laws [27]. Therefore, it is imperative to develop a mathematical model to estimate the uncertainty in InSAR deformation monitoring. In particular, the normalized difference vegetation index (NDVI) has been shown to effectively estimate the coherence level [28], [42], [43]. Consequently, if a quantitative functional model linking coherence and precision of deformation monitoring can be developed, NDVI could be directly used to estimate the precision of InSAR-derived deformation measurements, thus unlocking the full-application potential of InSAR technology.

To this end, a comprehensive experimental framework was developed to assess the uncertainty of InSAR-based surface deformation monitoring. Based on this framework, the InSAR deformation monitoring uncertainty (In-DMU) model was proposed for a priori estimation of deformation monitoring precision. Experimental results of multiplatform SAR datasets (ALOS-1, ALOS-2, Radarsat-2, Sentinel-1, and TerraSAR-X) demonstrated that the proposed In-DMU model can effectively estimate InSAR monitoring precision, which can provide valuable guidance for SAR data selection and multilooking configuration optimization. The contributions of this study are as follows.

1) Through multifactor controlled experiments, the combined effects of coherence, surface deformation, and SAR satellite parameters (i.e., wavelength and resolution) on monitoring precision were systematically quantified.
2) The In-DMU model was developed to mathematically formalize the synergistic dependencies among key influencing factors. In addition, its applicability boundary was defined using a change-point detection algorithm, namely, the Pettitt test based on the slidin-window standard deviation (Pt-SWStd).
3) The effectiveness and generalizability of In-DMU was validated using multiplatform SAR datasets in combination with in situ measurements.

The structure of this article is as follows. Section II describes the methodological framework used. Section III details the simulation experimental configuration. Section IV systematically analyzes the simulation results and establishes the In-DMU model. Section V validates the reliability of the In-DMU model. Section VI summarizes the key findings, discusses its limitations, and outlines future research directions.

II. Method

A. Simulation Experimental Method

The methodological framework of this study is outlined in Fig. 1. The simulation framework comprises three main stages: differential interferometric phase generation, phase processing, and result analysis and modeling. During the phase generation stage, the process involves: 1) constructing a high-spatial-resolution deformation field based on predefined deformation models; 2) resampling the deformation field to match the target resolution; 3) generating differential interferometric phases using the designed wavelength parameters (L, C, and X bands); and 4) applying a phase noise model to simulate interferometric phases under varying coherence conditions. In the phase processing stage, the minimum cost flow (MCF) algorithm [44] is employed to unwrap the interferometric phase, which

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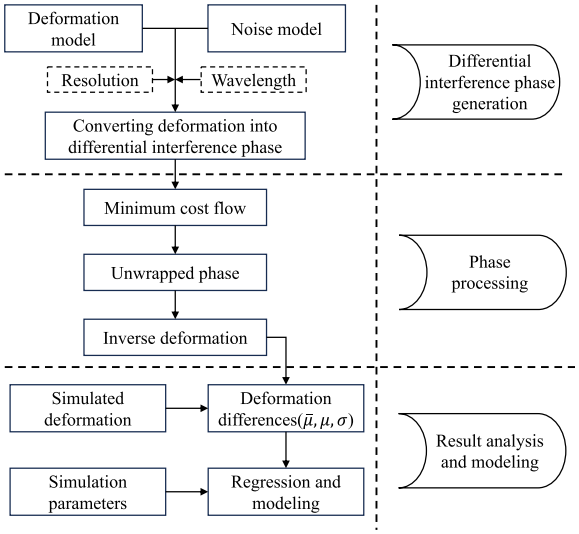


Fig. 1. Design of the methodological framework.

is subsequently converted into surface deformation. In the analysis and modeling stage, the simulated and recovered deformations are compared using quantitative metrics such as the root mean square error (RMSE) and STD. Finally, multivariate regression analysis [45] is employed to establish a quantitative relationship between the uncertainty of the deformation measurements and the key controlling parameters. In addition, the Pt-SWStd algorithm is used to identify the applicability boundary.

B. Phase Noise Model

The InSAR principle is based on the phase differencing between two SAR acquisitions of the same region to generate topographic maps or measure ground deformation [8]. However, InSAR-derived results are susceptible to multiple error sources [46]. Under the assumption of an independent error source, the total error σ_{total} can be expressed as follows [33]:

$$\sigma_{\text{total}}^2 = \sigma_{\text{sys}}^2 + \sigma_{\text{ran}}^2 + \sigma_{\text{loc}}^2 \tag{1}$$

where σ_{sys}, σ_{ran}, and σ_{loc} represent the systematic errors, random errors, and location-induced errors, respectively. The systematic errors include flat-earth phase, orbital residuals, topographic phase residuals, ionospheric delay, tropospheric delay, liquid water delay, scatterer phase shift, and surface movement-induced errors. These errors can be effectively mitigated through targeted mitigation strategies [47]. The inherent slant-range imaging geometry of SAR systems causes geometric distortions, including foreshortening, layover, and shadowing, as topographic features are projected to erroneous pixel locations, resulting in location-induced errors [48]. Geometric distortions are inevitable and are correlated with local topography [49]. Thus, location-induced errors on interferogram uncertainty are not considered in this study. The focus is on the uncertainty of single interferograms, for which only random errors are taken into account. Systematic errors (e.g., topographic phase residuals, ionospheric delay, and tropospheric delay) will be addressed in future work in conjunction with their impact on the uncertainty of time-series InSAR.

Decorrelation is a primary source of random errors [33]. Coherence γ serves as a crucial metric for quantifying phase noise levels and assessing interferogram quality, defined as follows [50]:

$$\gamma = \frac{E\left\{y_1 y_2^*\right\}}{\sqrt{E\left\{|y_1|^2\right\} E\left\{|y_2|^2\right\}}} \tag{2}$$

where γ is the coherence, with its magnitude bounded in the range $[0, 1]$. $E\{\cdot\}$ denotes the statistical expectation operator, y_1 and y_2 represent the registered complex SAR images, and $*$ denotes the complex conjugate operation. Coherence quantifies the correlation between two SAR images and reflects the spatiotemporal stability of surface scatterers' backscattering properties [13], [51]. Furthermore, coherence serves as the dominant parameter that governs the probability density function (pdf) of the interferometric phase, which can be formulated as follows [38], [41]:

$$\begin{aligned}&\text{pdf}(\varphi, \gamma, L, \varphi_0)\\ &= \frac{\Gamma(L+1/2)\left(1-\gamma^2\right)^L |\gamma| \cos(\varphi-\varphi_0)}{2\sqrt{\pi}\Gamma(L)\left(1-\gamma^2\cos^2(\varphi-\varphi_0)\right)^{L+1/2}}\\ &\quad + \frac{\left(1-\gamma^2\right)^L}{2\pi} F\left(L, 1; \frac{1}{2}; \gamma^2\cos^2(\varphi-\varphi_0)\right)\end{aligned} \tag{3}$$

where L is the multilook level, φ and φ_0 represent the interferometric phase and the expected interferometric phase, respectively, and $\Gamma(\cdot)$ and $F(\cdot)$ denote the Gamma function and the Gaussian hypergeometric function, respectively. Moreover, the interferometric phase standard σ_φ can be derived as follows [52]:

$$\sigma_\varphi^2 = \int_{-\pi}^{\pi} \left[\varphi - E\{\varphi\}\right] \text{pdf}(\varphi)\, d\varphi. \tag{4}$$

The interferometric phase STD under different parameter combinations can be computed using (4), as shown in Fig. S1 in the Supplementary Material. The phase standard asymptotically approaches approximately 1.81 rad as coherence approaches 0, while it converge to 0 when coherence reaches its maximum (i.e., 1). Moreover, a significant negative correlation is observed between the phase standard and the multilook level at fixed coherence values. Therefore, in practical applications, multilook processing effectively suppresses interferometric phase noise through spatial averaging, albeit at the cost of reduced spatial resolution [38]. However, while (4) quantifies interferometric phase noise, it cannot directly estimate InSAR deformation monitoring uncertainty, as it does not account for errors introduced during InSAR processing workflows (e.g., phase unwrapping) and other factors (see details in [27]). Notably, the phase noise model is constructed based on unbiased coherence, therefore, the models developed in this study are also established under the same assumption.

C. Change-Point Detection Algorithm

In sequential data, a data point is defined as a change point if the statistical properties of its pre- and post-subsequences

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TABLE I
CONFIGURATION OF SIMULATION EXPERIMENTAL PARAMETERS

Parameter	Value
Wavelength (mm)	235, 55.5, 31.1
Resolutions (m)	1×1, 2×2, 3×3, 4×4, 5×5, 6×6, 8×8, 10×10, 12×12, 16×16, 20×20, 24×24
Coherence	0:0.01:1
Deformation (mm)	0, 5, 10, 20, 30, 40, 50, 60, 80, 100, 120, 140, 160, 180, 200, 250, 300, 350, 400, 500, 600, 700, 800, 900, 1000, 1200, 1400, 1600, 1800, 2000, 2500, 3000, 3500, 4000
Number of parameter combinations	123,624

differ significantly. Numerous change point detection algorithms have been proposed, including Bayesian inference [53], the Mann–Kendall test [54], the moving t-test [55], Yamamoto's method [56], Pettitt's test [57], and the Bernaola-Galvan segmentation algorithm [58], to automatically identify the locations of change points. These techniques have been extensively validated and widely applied in geoscience and environmental monitoring [59], [60]. In this study, a Pettitt test integrated with a sliding-window standard deviation (Pt-SWStd) is proposed for the precise detection of change points in the simulated experimental results. The procedure is as follows.

First, the sliding-window STD algorithm is used to quantify the dispersion of data within sequential segments [61]. Specifically, for a data sequence $\{y_1, y_2, y_3, \ldots, y_m\}$ of length m, a sliding window of size n ($n \ll m$) is initialized at the sequence origin and progressively advanced with single-step increments. At each step, the STD (σ) within the current window is calculated. Subsequently, the Pettitt test [56], a robust nonparametric method based on the rank sequence analysis, is employed applied to the resulting STD sequence generated by the sliding-window algorithm to automatically detect change points.

III. DATA SIMULATION AND RESULTS

A. Simulation Experimental Configuration

In this study, a numerical simulation framework has been established to evaluate the uncertainty of deformation measurements. The multiparameter control experiment with a complete factorial combination [62] was conducted to systematically assess the performance of different combinations of parameters. The experiments have taken into account wavelengths, spatial resolutions, coherence levels, and deformation magnitudes. Specifically, the experimental configuration includes three wavelengths (L/C/X bands), 12 spatial resolutions (1–24 m) and 101 coherence levels (0–1), 34 deformation magnitudes (5–4000 mm), generating a total of 123 624 parameter combinations (see Table I). Each trial was repeated 100 times, with statistical averaging applied to yield robust estimation results.

B. Simulation Interferometric Phase Results

Figs. 2 and 3 present the representative differential interferogram patches to elucidate the coupling mechanisms between wavelength, spatial resolution, coherence, and deformation magnitude on interferometric phase characteristics. Fig. 2 shows the X band (λ = 31.1 mm) differential interferometric patches with deformation magnitudes of 10, 50, 100, 400, 1000, and 4000 mm, spatial resolutions of 1 × 1, 5 × 5, and 20 × 20 m, and coherence levels of 0.4 and 0.8. It can be seen from Fig. 2 that lower coherence results in increased phase noise, progressively blurring interferometric fringes. At the same coherence level, a lower spatial resolution (larger pixel size) decreases the fringe clarity. This highlights the importance of high coherence and high resolution to achieve high precision deformation retrieval. For example, in interferogram patches with a resolution of 20 × 20 m and a deformation magnitude of 400 mm: at γ = 0.4, the interferometric phase fringes appear blurred and are ineffective for monitoring, whereas at γ = 0.8, the fringes are clear, enabling effective deformation estimation. In addition, minor deformations can cause spectral aliasing between the deformation and noise phases, while large deformations can lead to fringe densities exceeding Nyquist sampling limits [63]. Both cases introduce phase-unwrapping errors in surface deformation inversion [33].

Fig. 3 presents differential interferometric patches with a spatial resolution of 20 × 20 m, deformation magnitudes of 10, 50, 100, 400, 1000, and 4000 mm; wavelengths of 31.1, 55.5, and 235 mm; and coherence levels of 0.4 and 0.8. An inverse proportionality between the wavelength and the fringe density is observed for equivalent deformation magnitudes, as shown in Fig. 3. This indicates that long-wavelength systems can accommodate larger deformation magnitudes without phase aliasing. However, their lower phase sensitivity limits their ability to detect small deformations compared to that of short-wavelength systems.

In summary, the uncertainty of InSAR deformation monitoring is influenced by coherence, resolution, wavelength, and deformation magnitude. High-coherence and high-resolution SAR images are more effective for deformation monitoring, while selecting the optimal SAR wavelength requires careful consideration of both ground deformation and noise conditions.

IV. CONSTRUCTION OF FUNCTIONAL MODELS

A. Analysis of Deformation Inversion Results

Fig. 4 illustrates the flowchart of the simulation result analysis and modeling process. First, using the simulated surface deformation fields as a reference, RMSE and STD of the retrieved deformation were calculated for all experimental groups. The typical RMSE results are shown in the Supplementary of Fig. S2, and the typical STD results are presented in Fig. 5. The results, as shown in Fig. 5 are consistent with the Nyquist–Shannon sampling theorem [63], [64], thus validating the reliability of the simulation tests. Moreover, these results demonstrate the coupled effects of deformation magnitude, radar wavelength, spatial resolution, and coherence on the uncertainty of InSAR-based deformation monitoring, as detailed as follows.

1) When the deformation gradient is significantly below the critical detection threshold (i.e., $\lambda/2R$), the

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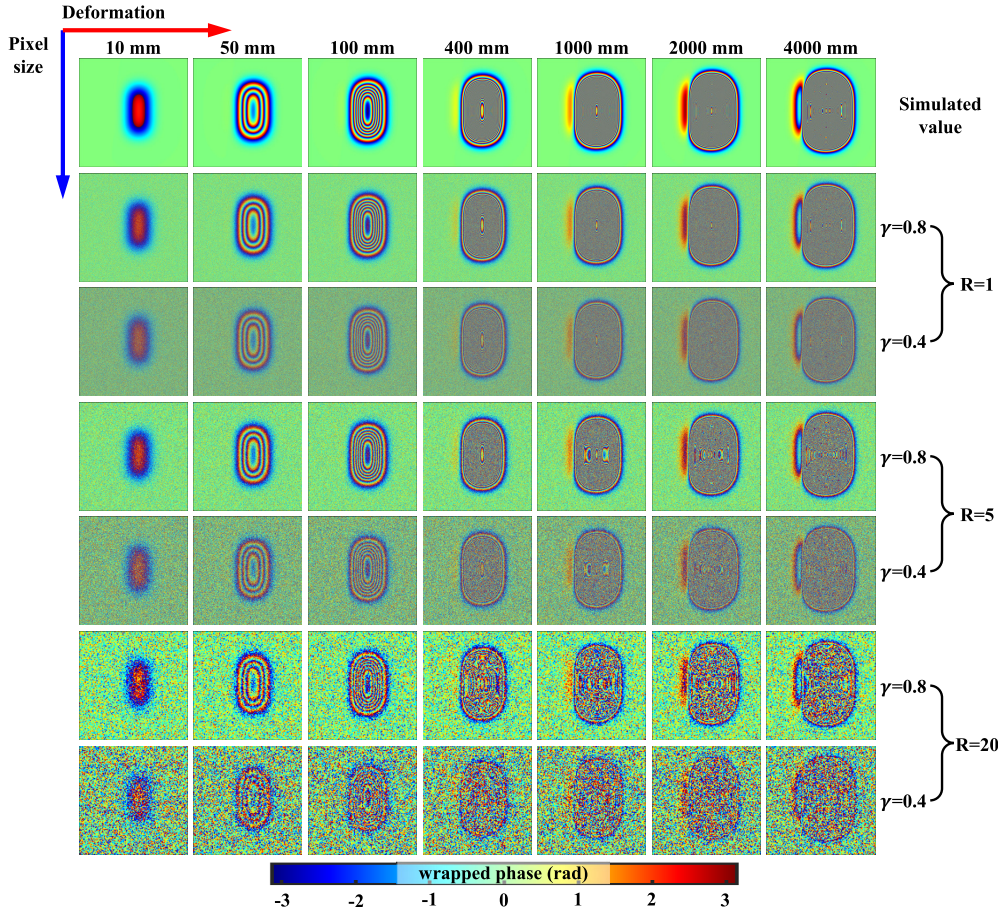


Fig. 2. Differential interferogram patches corresponding to deformation magnitudes 10-, 50-, 100-, 400-, 1000-, and 4000-mm, resolutions $R = 1 \times 1$, 5×5, and 20×20 m and coherence $\gamma = 0.4$ and 0.8, respectively. The spatial extent of each of the interferogram patches is 2500×2500 m, and the wavelength λ is 31.1 mm.

distributions of RMSE and STD are primarily influenced by coherence, with only weak dependence on other parameters, as shown in the top two rows subfigures of Fig. 5. This suggests that phase noise is the dominant source of uncertainty in deformation monitoring under this condition.

2) As the deformation magnitude increases, the distribution of uncertainty in deformation monitoring exhibits clear and systematic trends, becoming increasingly sensitive to spatial resolution and radar wavelength. This is evidenced by the significant differences observed between low- and high-resolution data in Fig. 5(d)–(e), (h)–(i), and (m)–(n).
3) When deformation gradients exceed the critical threshold, both RMSE and STD remain significantly above 0, even under perfect coherence ($\gamma = 1$), as shown in Fig. 5(j) and (o). This phenomenon primarily results from the failure to accurately resolve the integer ambiguity during the phase-unwrapping process, leading to the breakdown of the deformation estimation [63].
4) In terms of wavelength, short-wavelength data generally yield higher monitoring precision due to greater phase sensitivity, assuming equal coherence. In fact, shorter wavelengths are more vulnerable to coherence loss in real-world scenarios, which compromises the accuracy of deformation measurements [65]. Moreover, short wavelengths are more susceptible to limitations imposed by the Nyquist–Shannon sampling theorem [66]. In contrast, longer wavelength data show more stable performance with increasing deformation, making them better suited for monitoring large deformations, as illustrated in the second and third row subfigures of Fig. 5.

The monitoring results are classified into three distinct regimes according to the RMSE and STD distribution characteristics: the wavelength–coherence dominant regime, the transition regime, and the retrieval breakdown regime. The first type of error distribution sequence features a single change point, where the low-coherence segment preceding the change point follows a linear trend, while the high-coherence segment after the change point exhibits a nonlinear trend. The second type also shows an evident change point, but unlike the first type, it exhibits an "S-shaped" nonlinear trend in regions with low coherence. The third type lacks identifiable change points, with errors remaining nearly constant or forming a partial inverse "S-shaped" curve. This corresponds to deformation magnitudes exceeding InSAR's theoretical detection limits, making the inversion results unreliable. In this study, the first

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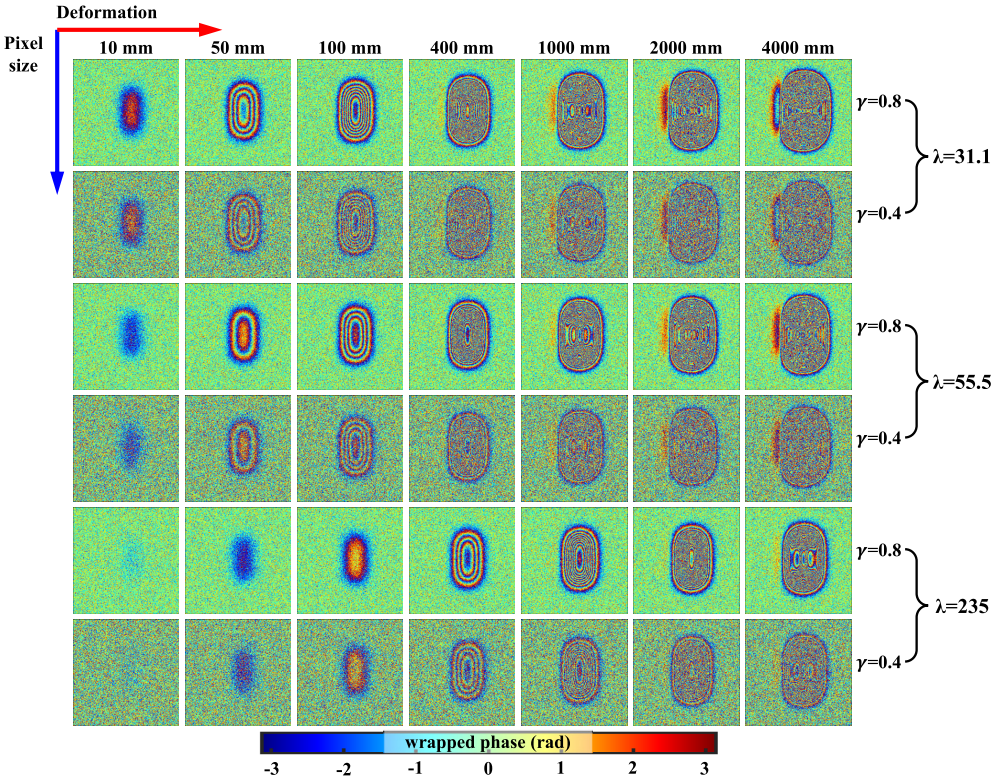


Fig. 3. Differential interferogram patches corresponding to deformation magnitudes 10-, 50-, 100-, 400-, 1000-, and 4000-mm, wavelengths λ = 31.1, 55.5, and 235 mm and coherence γ = 0.4 and 0.8, respectively. The spatial extent of each of the interferogram patches is 2500 × 2500 m, and the resolution R is 20 m.

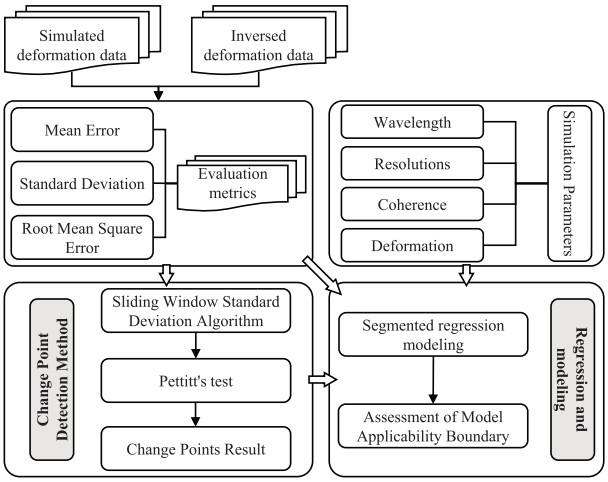


Fig. 4. Flowchart of the simulation result analysis and modeling process.

error category was used to establish the InSAR deformation monitoring uncertainty (In-DMU) model, while the second category was used to define the applicability boundary of the constructed In-DMU model. The third category was excluded from the analysis due to the unreliability of the inversion results.

B. Modeling of Deformation Monitoring Uncertainty

To analyze the distribution of the monitoring errors, the mean error (ME) of the deformation estimates was calculated under the wavelength–coherence dominant regime, as shown in Fig. 6(a)–(c). Regardless of the resolution and wavelength, the ME exhibits a divergent trend centered around 0. This suggests a statistically significant negative correlation between the uncertainty of the deformation monitoring and the coherence levels. Subsequently, a regression analysis was performed to establish a quantitative relationship among ME, coherence γ, and resolution R

$$\mathrm{ME}_{\lambda=235\ mm}(\gamma, R) = 7.46 - 8.901\gamma - 0.01574R \quad (5)$$

$$\mathrm{ME}_{\lambda=55.5\ mm}(\gamma, R) = 0.7335 - 0.6259\gamma - 0.01793R \quad (6)$$

$$\mathrm{ME}_{\lambda=31.1\ mm}(\gamma R,) = -0.0926 + 0.0310\gamma + 0.0054R. \quad (7)$$

Equations (5)–(7) indicate that the fit planes derived from different wavelengths are nearly 0, and spatial resolution has a minimal impact on ME. Specifically, under identical coherence conditions, the maximum ME variation due to resolution differences is only 0.41 mm, i.e., $0.01793 \cdot (R_{\max} - R_{\min})$, suggesting that the effect of resolution on ME can be neglected. Fig. 6(d)–(f) shows the coherence-dependent distribution characteristics of ME, with 80.31% of samples clustering within $\pm(\lambda \cdot \sigma_\varphi)/4\pi$ range. Furthermore, when coherence drops below 0.7 (corresponding to an interferometric phase STD $\sigma_\varphi >$ 1.08 rad), the proportion of samples with ME in the range of $\pm(\lambda f5 \cdot \sigma_\varphi)/4\pi$ to $\pm(\lambda \cdot \sigma_\varphi)/2\pi$ gradually increases, with 25.34% of samples falling within this range.

From the top two rows of subfigures of Fig. 5 and Fig. S2 in the Supplementary Material, it can be observed that a monotonically increasing trend is exhibited in both STD and RMSE as decreasing coherence levels. In addition, change

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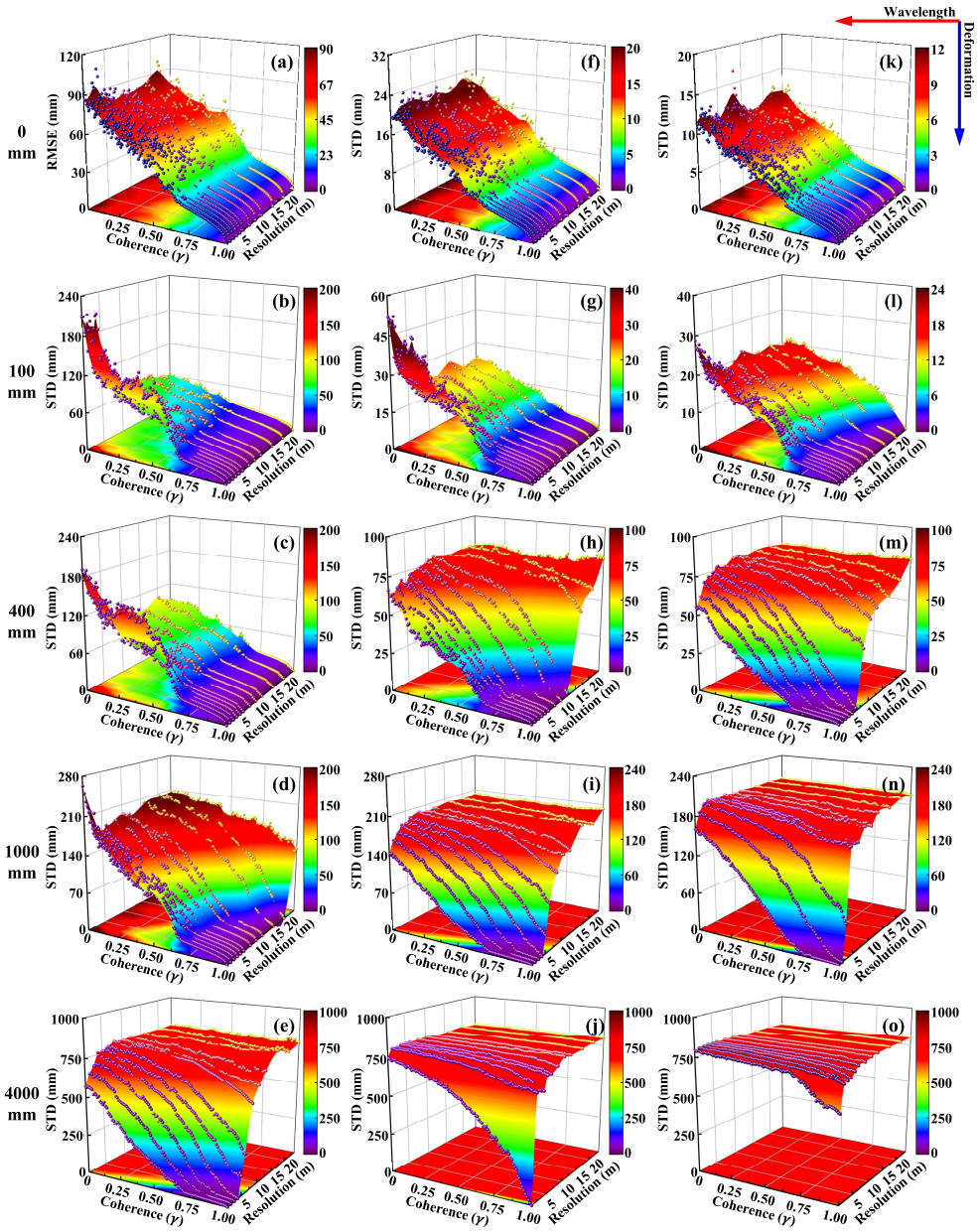


Fig. 5. (a)–(e) Distributions of STD at a wavelength of 235 mm for maximum deformations of 0, 100, 400, 1000, and 4000 mm, respectively. (f)–(j) and (k)–(o) Corresponding results for wavelengths of 55.5 and 31.1 mm, respectively.

points are present in both the RMSE and STD results, indicating significant differences in statistical characteristics before and after the change points. This characteristic is particularly pronounced in STD sequences. Therefore, the proposed Pt-SWStd method (see Section II-C) was applied to detect the change points based on the STD results, as detailed in Table S1 in the Supplementary Material.

The segmented regression method was employed to model the pre- and post-change point data series. Specifically, a linear regression model was applied to the prechange point segment (i.e., $0 < \gamma < \gamma_{cp}$), while an exponential function was used for the nonlinear fitting of the postchange point segment (i.e., $\gamma > \gamma_{cp}$). The regression functions are shown in (S1)–(S3) in the Supplementary Material.

It can be observed from (S1) to (S3) in the Supplementary Material that the coefficient preceding the spatial resolution term R is relatively small, with a maximum value of only 0.1002. This indicates that the influence of spatial resolution on the uncertainty of deformation monitoring is limited. Table II quantitatively demonstrates the impact of spatial resolution parameters on both RMSE and STD. It is evident that under the same coherence conditions, the maximum difference in RMSE due to spatial resolution is 1.97 mm, and the maximum difference in STD is 1.38 mm. This result indicates that the effect of spatial resolution on the uncertainty of the deformation measurements is negligible.

Disregarding resolution effects, regression models were developed to explore the relationships among RMSE, STD, and coherence, as shown in Fig. 7. In postchange point scenarios, both the RMSE and STD regression models exhibit excellent goodness-of-fit ($R^2 > 0.93$). In prechange point scenarios, the RMSE regression model achieves an R^2

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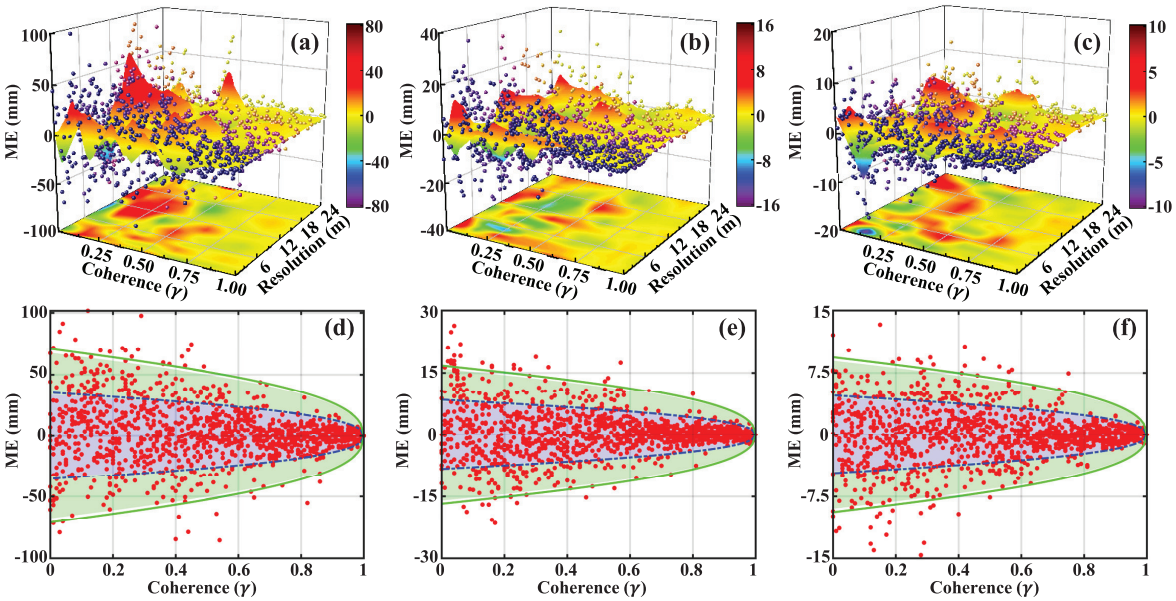


Fig. 6. (a)–(c) 3-D distributions of ME as functions of coherence and resolution under distinct wavelengths (235, 55.5, and 31.1 mm, respectively) in the 0-mm deformation scenario. (d)–(f) Distributions of resolution-independent MEs for corresponding experimental groups. The light blue and dark green regions in (d)-(f) denote $\pm(\lambda \cdot \sigma_\varphi)/4\pi$ and $\pm(\lambda \cdot \sigma_\varphi)/2\pi$ error bounds, respectively.

TABLE II

MAXIMUM DIFFERENCES IN RMSE AND STD AT THE SAME SPATIAL RESOLUTION LEVELS

Wavelength (mm)	235		55.5		31.1	
Segment	$0 \le \gamma \le 0.64$	$\gamma > 0.64$	$0 \le \gamma \le 0.66$	$\gamma > 0.66$	$0 \le \gamma \le 0.65$	$\gamma > 0.65$
RMSE (mm)	1.61	0.22	1.71	0.88	1.97	0.81
STD (mm)	1.38	0.49	0.37	0.11	0.13	0.46

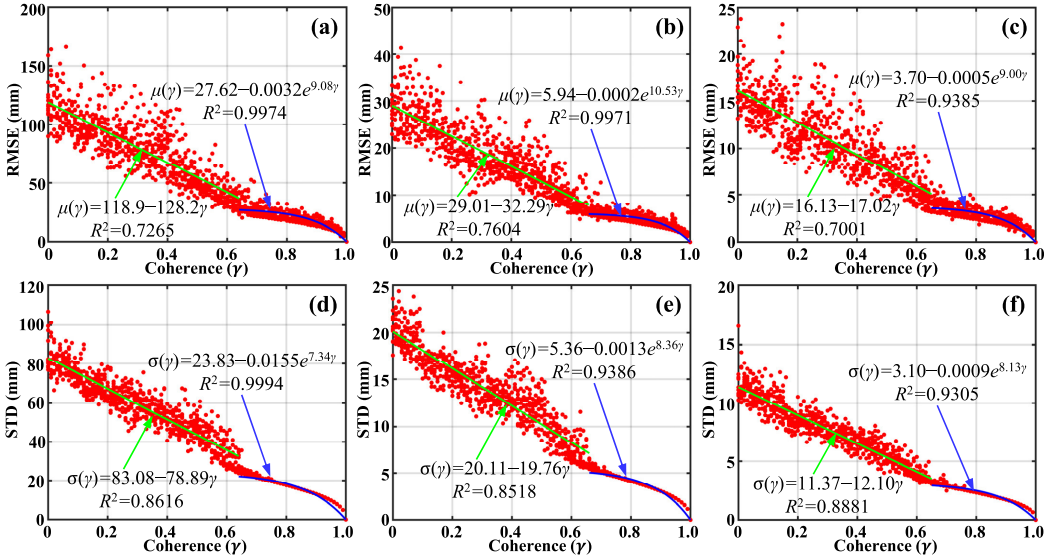


Fig. 7. (a)–(c) Regression analysis results of RMSE as a function of coherence for simulated experiments with wavelengths of 235, 55.5, and 31.1 mm, respectively (resolution effects disregarded). (d)–(f) Regression analysis results of STD as a function of coherence for the corresponding experimental groups.

value above 0.7, while the STD regression model exceeds 0.85. These results suggest that the resolution-independent function models constructed can effectively estimate the reliability (RMSE and STD) of the deformation inversion results.

To elucidate the coupling mechanisms between coherence and wavelength in the uncertainty of deformation monitoring, a multivariate regression analysis was performed to establish quantitative relationships among coherence (γ), wavelength (λ), and precision metrics (RMSE and STD),

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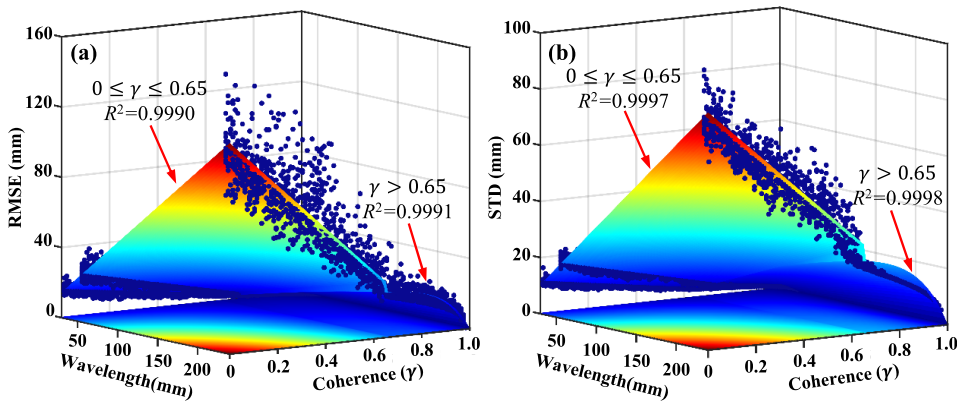


Fig. 8. Multivariate regression results of (a) coherence–wavelength–RMSE and (b) coherence–wavelength–STD.

as shown in (8) and Fig. 8. In this process, $\gamma = 0.65$, representing the mean value of the detected change points, was used as the overall change point. The regression models exhibit exceptional goodness-of-fit, with all R^2 values exceeding 0.999. These results indicate that the models constructed effectively describe the relationships among these parameters.

$$\mathrm{RMSE}(\gamma,\lambda)=\begin{cases}0.00079e^{\frac{11.4\lambda+496}{\lambda+70.24}}\\ \quad -0.00079\ e^{\frac{11.4\lambda+496}{\lambda\gamma+70.24}},\gamma>0.65\\ -0.5145+0.503\lambda+1.282\gamma\\ \quad -0.5713\lambda\gamma,0\le\gamma\le 0.65\end{cases}$$

$$\mathrm{STD}(\gamma,\lambda)=\begin{cases}0.0066e^{\frac{8.992\lambda+260.3}{\lambda+56.76}}\\ \quad -0.0066\ e^{\frac{8.992\lambda+260.3}{\lambda\gamma+56.76}},\gamma>0.65\\ -0.192+0.3538\lambda-1.384\gamma\\ \quad -0.3334\lambda\gamma,0\le\gamma\le 0.65\end{cases}\tag{8}$$

As shown in (8), the constructed uncertainty estimation model includes two indicators, i.e., RMSE(γ,λ) and STD(γ,λ). In practical applications, STD is commonly used to represent the precision of deformation measurements and maintains a convertible relationship with RMSE [67]. Therefore, STD(γ,λ) is used to estimate the precision of the InSAR deformation measurements, which serves as the In-DMU model. In addition, the In-DMU model demonstrates a negative correlation between wavelength and deformation monitoring precision. This is primarily due to wavelength acting as a scaling factor in the phase-to-deformation conversion process (i.e., $\lambda\cdot\varphi/4\pi$), which explains why longer wavelengths result in higher monitoring errors under equivalent noise conditions [22]. It should be noted that long-wavelength SAR data typically exhibit better coherence than shorter wavelengths under identical ground conditions [65]. Therefore, it is essential to comprehensively consider multiple factors when selecting the optimal wavelength parameters.

C. Assessment of the Model Applicability Boundary

To determine the applicability boundary of the constructed In-DMU model, the Pt-SWStd algorithm was applied to detect change points in different experimental groups, as shown in Fig. 9. It is observed that the precision of deformation measurement is governed by the synergistic effects of interferometric phase noise and deformation gradient. Specifically, when the deformation magnitudes are small, coherence-related phase noise is the primary factor affecting monitoring uncertainty, and the detected change points are minimally impacted by resolution. As the deformation magnitude increases, the deformation gradients become more significant, eventually emerging as the primary factor influencing the precision of the InSAR-derived deformation. In addition, spatial resolution plays a crucial role in determining deformation gradients—a coarser resolution results in larger deformation gradients for the same deformation field. As the magnitude of the deformation increases, the change points shift toward higher coherence, and the migration rate increases more rapidly at coarser resolutions, as shown in Fig. 9.

Fig. 10 presents the distribution of the change point relative to the difference between the critical deformation gradient ($G_{\mathrm{cri}}=\lambda/2R$) and the simulated deformation gradient (D_{sim}), defined as $g = G_{\mathrm{cri}} - G_{\mathrm{sim}}$. An exponential decay model ($a\cdot\exp(-b\cdot x)+c$) was used to fit the distribution (Fig. 10), yielding R^2 values of 0.924, 0.901, and 0.836 for the L band ($\lambda = 235$ mm), the C band ($\lambda = 55.5$ mm), and the X band ($\lambda = 31.1$ mm), respectively. These results indicate that the exponential decay function effectively describes the distribution pattern.

When the deformation gradient is significantly smaller than the critical deformation gradient (i.e., $g \to \lambda/2R$), the coherence of change points asymptotically approaches specific constants. Specifically, for wavelengths of 235, 55.5, and 31.1 mm, the coherence values stabilize at 0.68, 0.68, and 0.69, respectively. These values are close to the overall coherence of the change point of 0.65. In addition, setting a coherence variation threshold of $\Delta\gamma = 0.01$, which is determined based on the coherence estimation accuracy and detectability [68], [69], the gradient difference thresholds are derived for the three bands, with values of 6.433 (235 mm), 1.407 (55.5 mm), and 0.456 (31.1 mm). This suggests that when the deformation gradient meets these conditions, the In-DMU model can reliably estimate the precision of the deformation measurements.

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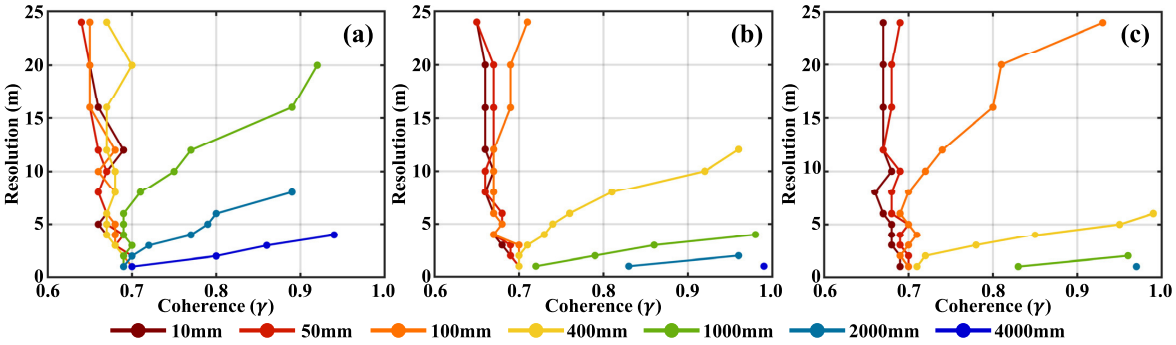


Fig. 9. Change point detection results in deformation scenarios. (a)–(c) Wavelengths of 235, 55.5, and 31.1 mm, respectively.

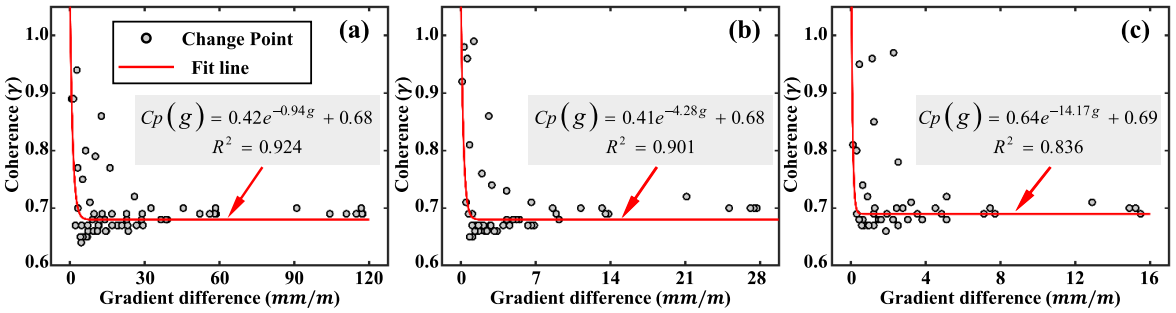


Fig. 10. Distribution of change points relative to the difference between detectable critical gradient thresholds and simulated maximum deformation gradients. (a)–(c) Wavelengths of 235, 55.5, and 31.1 mm, respectively.

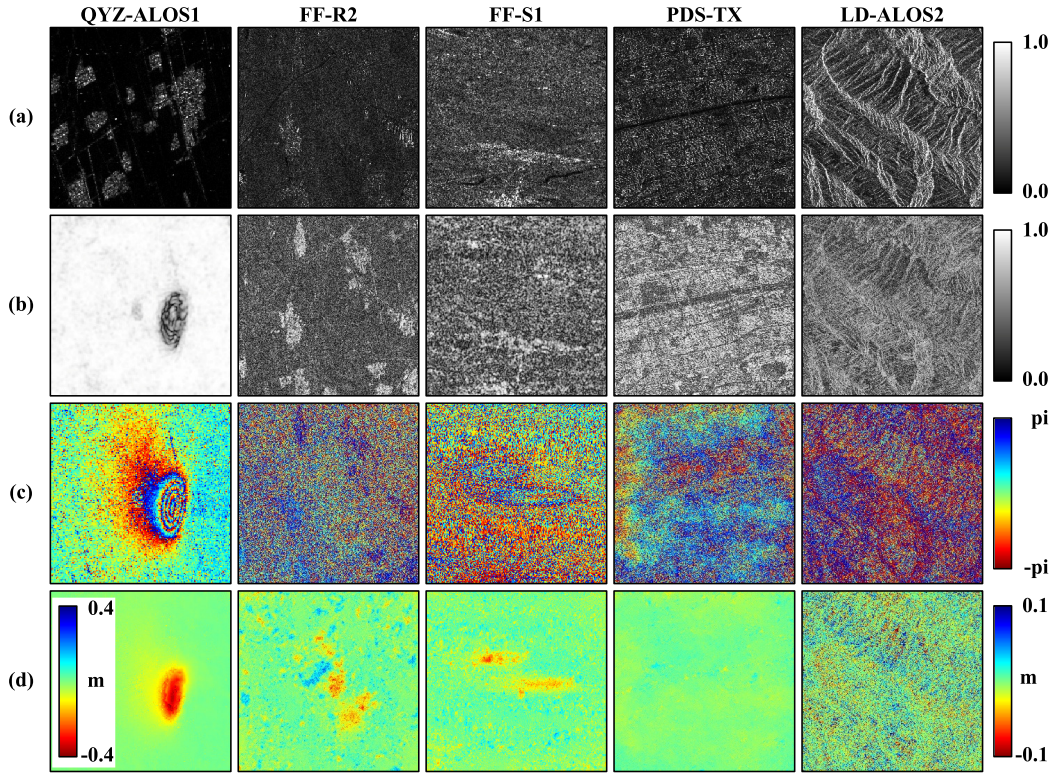


Fig. 11. Characterization of the study area. (a) SAR intensity maps. (b) Coherence coefficient maps. (c) Differential interferograms. (d) LOS deformation displacement fields.

The synthesized findings demonstrate that when the maximum deformation gradient within a region is below the defined threshold, the precision of deformation measurements is predominantly controlled by wavelength and coherence level,

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TABLE III
OVERVIEW OF THE STUDY AREAS AND SAR DATASETS

Study areas	SAR type	Reference data	Secondary data	Wavelength	Resolution (m) $R \times A$	Max deformation gradient (mm/m)
QYZ	ALOS-1	2010/01/13	2010/02/28	L-band (235 mm)	4.68×3.17	2.37
FF	Radarsat-2	1015/11/30	2015/12/24	C-band (55.5 mm)	2.66×2.91	0.78
	Sentinel-1	2015/12/02	2015/12/26	C-band (55.5 mm)	2.33×13.94	1.38
PDS	Terrsar-X	2017/06/12	2017/06/23	X-band (31.1 mm)	0.93×2.11	-
LD	ALOS-2	2019/11/04	2019/12/30	L-band (235 mm)	1.43×2.13	-

"R" and "A" denote the range resolution and azimuth resolution, respectively.

while remaining independent of spatial resolution. Furthermore, the In-DMU model established in this study effectively estimates theoretical monitoring precision. It is important to note that multilooking processing was not implemented in the simulation experiments. Multilook processing can reduce interferometric phase noise, thus enhancing the monitoring precision, as shown in (3). To facilitate practical applications, this study also provides: 1) a deformation monitoring precision lookup table for various coherence–wavelength combinations (Table S2 in the Supplementary Material); 2) a maximum detectable critical deformation gradient table across wavelength–resolution combinations (Table S3 in the Supplementary Material); and 3) a relationship table between multilooking levels and equivalent coherence values (Table S4 in the Supplementary Material).

V. MODEL VALIDATION

Four representative land cover regions (see Fig. 11) were selected to validate the performance of the proposed In-DMU model, using SAR datasets from five typical SAR satellites (ALOS-1, ALOS-2, Sentinel-1, Radarsat-2, and TerraSAR-X), as detailed in Table III. Qianyingzi (QYZ) and Fengfeng (FF) are mining regions experiencing significant surface deformation, while Pingdingshan (PDS) and Loudi (LD) represent the urban and mountainous areas, respectively, with negligible deformation during the monitoring period, as shown in Fig. 11(a). Furthermore, to ensure the applicability of the constructed In-DMU model to the FF and QYZ mining areas, the maximum deformation gradient during the InSAR monitoring period was estimated using the probability integral method (PIM) [8], as shown in Table III. Specifically, geometric parameters of the underground working faces and PIM parameters were obtained [25], [70] for the monitoring period to calculate the 3-D surface deformation field via PIM. The PIM-derived deformation fields were then projected onto the line-of-sight (LOS) direction based on the SAR satellite imaging geometry (i.e., incidence and azimuth angles), and subsequently transformed into the SAR reference frame to derive the theoretical maximum deformation gradient. The results indicate that the In-DMU model applies to both FF and QYZ. The estimated deformation results using PIM are used solely to evaluate the applicability of the proposed model and are not included in subsequent analyses. Notably, the proposed model is developed based on the noise characteristics of the interferograms and is applicable to both ascending and descending SAR datasets. However, in the regions with available in situ measurements (QYZ and FF), only ascending-pass data are accessible. Although no ground-truth data are available for the LD and PDS areas, the deformation patterns in these regions are well understood by the authors, and only ascending-pass data are available there as well. Therefore, all SAR datasets used in this study are from ascending orbits.

The dual-pass differential interferometric synthetic aperture radar (D-InSAR) technique was used for surface deformation retrieval. First, precise coregistration of the selected SAR datasets was achieved through amplitude cross-correlation optimization. The differential interferograms were then formed by performing complex conjugate multiplication in the coregistered image pairs, followed by computing the coherence level [Fig. 11(b)]. The 30-m digital elevation model (DEM) from the Shuttle Radar Topography Mission (SRTM) [71] was used to remove the topographic phase, resulting in the differential interferogram phase [see Fig. 11(c)]. The MCF algorithm was applied to unwrap the interferometric phase, enabling the derivation of LOS deformation fields [see Fig. 11(d)]. In this study, a 5×5 pixel estimation window was used for the coherence calculation.

To validate the reliability of the proposed model, leveling measurements temporally synchronized with InSAR monitoring were collected from 35 stations in QYZ and 71 stations in FF, as shown in Fig. 12. The differences between InSAR-derived deformations and leveling measurements, termed ME, were calculated. In addition, the In-DMU model was applied to estimate the error at each station, called IEE. It should be noted that both QYZ and FF represent mining-induced deformation areas, where vertical subsidence is significantly greater than horizontal displacement [14]. Furthermore, due to the relatively small incidence angle of the SAR satellite, vertical motion predominantly contributes to the LOS displacement in the 3-D deformation field [14]. This implies that the influence of horizontal displacement on LOS measurements can be reasonably neglected in these areas. Therefore, leveling measurements were projected onto the LOS direction using the geometric mapping relationship to derive the monitoring errors. This simplification does not introduce significant systematic errors and has a negligible impact on the final results. Furthermore, coherence-based errors, called CEEs, were estimated using phase noise combined with the error propagation law [27]. It can be seen from Fig. 12 that the InSAR-derived deformations are highly consistent with the leveling measurements. Moreover, CEEs are significantly smaller than the actual measurement errors, with corresponding linear fit

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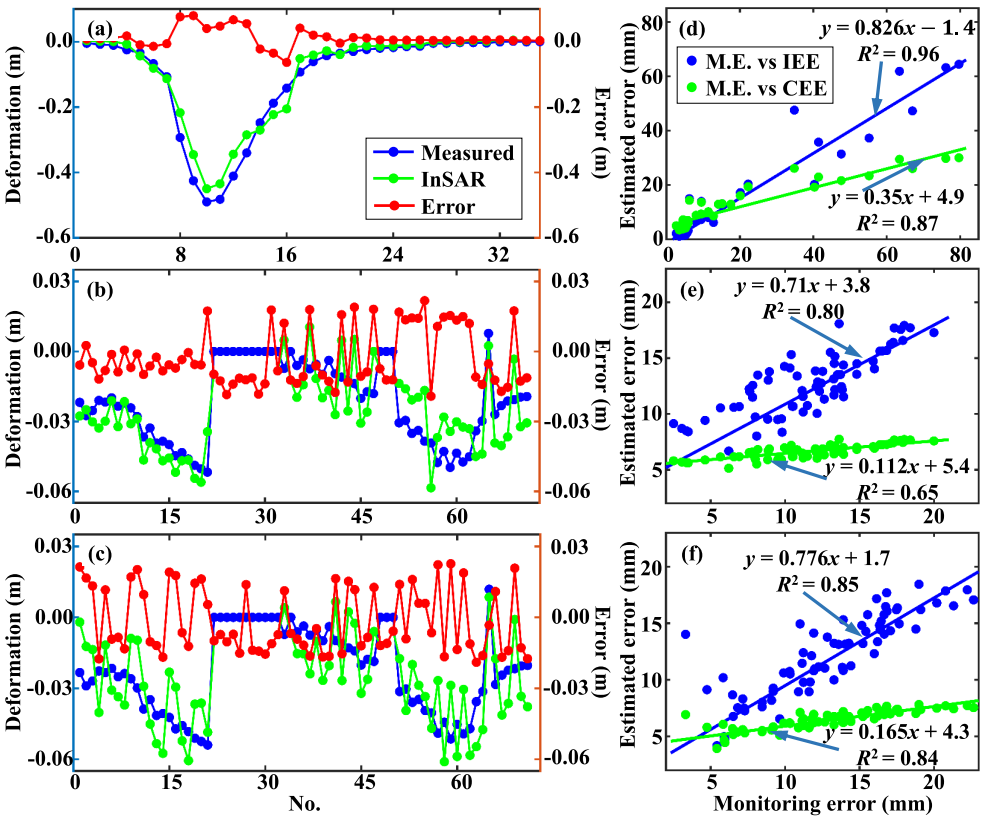


Fig. 12. (a)–(c) Comparative analysis between InSAR-derived deformations and leveling measurements for QYZ (ALOS-1) and FF Mining Areas (RADARSAT-2 and Sentinel-1). (d)–(f) Correlation between In-DMU estimated errors and monitoring errors for the corresponding study areas. "ME," "CEE," and "IEE" represent the monitoring error, the coherence-based estimated error, and the In-DMU estimated error, respectively.

TABLE IV
CORRELATION COEFFICIENT BETWEEN ESTIMATED AND ACTUAL ERRORS UNDER DIFFERENT WINDOW SIZES AND COHERENCE LEVELS

Group	$\gamma \geq 0.6$					$\gamma < 0.6$				
Window size	3×3	5×5	7×7	9×9	11×11	3×3	5×5	7×7	9×9	11×11
R^2	0.62	0.90	0.63	0.78	0.81	0.52	0.80	0.62	0.40	0.30

slopes of 0.35, 0.112, and 0.165, respectively. This indicates that the monitoring precision is significantly overestimated when directly inferred from phase noise. Fortunately, the proposed In-DMU model effectively mitigates this overestimation phenomenon, with corresponding fit line slopes of 0.826, 0.71, and 0.776, slightly lower than 1. These results indicate that the constructed model can effectively estimate the precision of the InSAR deformation measurements.

In practical applications, accurately obtaining unbiased coherence is challenging due to factors such as estimation window size, the heterogeneity of SAR pixels within the window, and the intrinsic bias of coherence estimators. Abdelfattah and Nicolas [68] and Touzi et al. [69] demonstrated that when the true coherence magnitude exceeds 0.6, the estimation bias is typically less than 0.02, allowing the estimate to be effectively considered impartial. However, when the true coherence magnitude falls below 0.6, the estimation bias becomes nonnegligible. Therefore, to evaluate the impact of coherence bias on the performance of the In-DMU model, the correlation coefficient R^2 between the model-estimated and actual monitoring errors was calculated by aggregating three datasets (QYZ-ALOS-1, FF-RADARSAT-2, and FF-Sentinel-1), grouping the data based on whether the estimated coherence magnitude was above or below 0.6, and distinguishing only by estimation window size, as listed in Table IV. In addition, the estimation and monitoring errors across various window sizes for the three datasets are compared, as shown in Fig. 13.

As shown in Table IV, when the coherence magnitude exceeds 0.6, the maximum and minimum correlation coefficients between estimation and monitoring errors are 0.90 and 0.62, respectively, with an average of 0.75. In contrast, for coherence values below 0.6, the maximum and minimum coefficients are 0.80 and 0.30, with an average of 0.53. These results indicate that the proposed In-DMU model can effectively estimate the uncertainty of deformation monitoring under high-coherence conditions (i.e., $\gamma \geq 0.6$), while still offering a meaningful reference value under low-coherence conditions (i.e., $\gamma < 0.6$). This highlights the critical importance of accurate coherence estimation in

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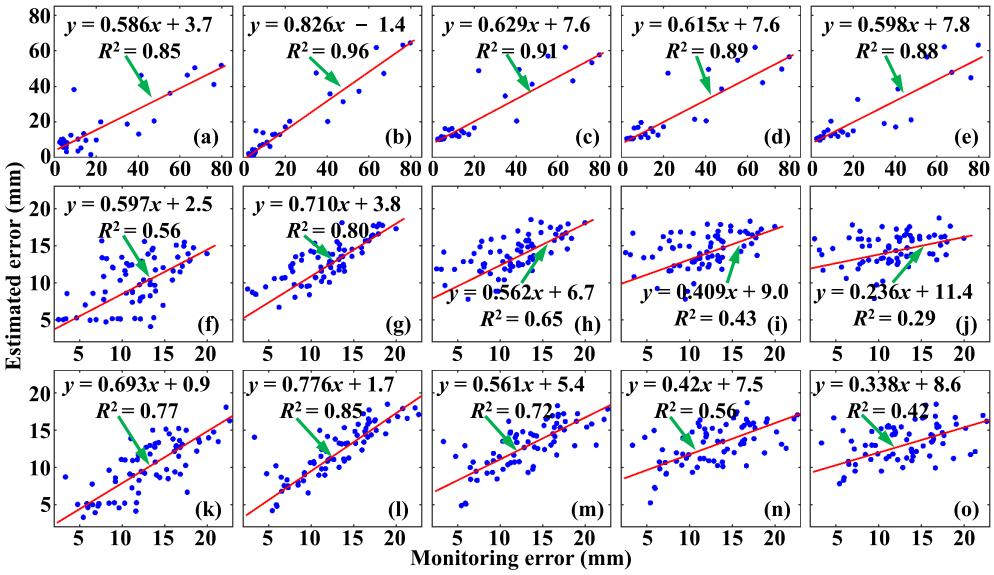


Fig. 13. (a)–(e) Correlation between monitoring errors and In-DMU estimated errors in the Qingyingzi (ALOS-1) area under different coherence estimation window sizes (i.e., 3 × 3, 5 × 5, 7 × 7, 9 × 9, and 11 × 11, respectively). (f)–(j) and (k)–(o) Corresponding results for the FF mining area using RADARSAT-2 and Sentinel-1 data, respectively.

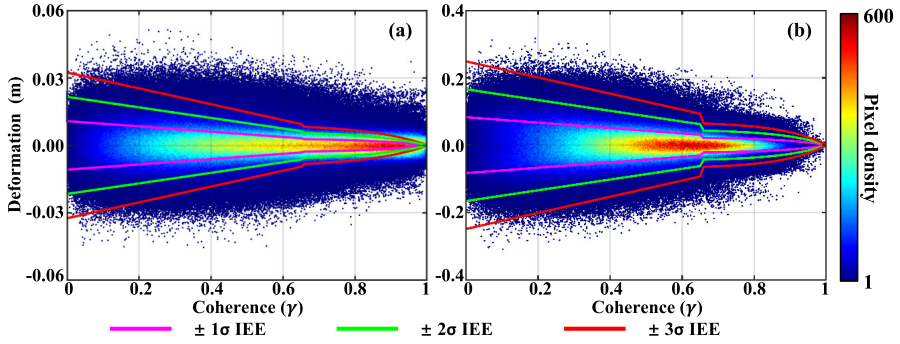


Fig. 14. Spatial characteristics of coherence coefficient and InSAR-derived deformation in (a) PDS and (b) LD areas. "IEE" represents the In-DMU estimated error.

ensuring the effectiveness of the model. In addition, it can be seen from Fig. 13 that as window size increases from 3 × 3 to 11 × 11, the fit line slopes and R^2 values for three datasets (QYZ-ALOS-1, FF-RADARSAT-2, and FF-Sentinel-1) exhibit nonmonotonic trends with initial increases followed by decreases. This suggests that moderate window enlargement can enhance coherence stability and reduce random noise, thereby improving the agreement between In-DMU estimated and observed errors. However, when the window size exceeds a certain threshold (e.g., 7 × 7, 9 × 9, or 11 × 11), the characterization performance of the In-DMU model begins to degrade. This phenomenon is likely attributed to spatial averaging, as oversized windows tend to oversmooth localized deformation features or heterogeneous noise distributions. Among all tested configurations, the 5 × 5 window exhibited the best overall performance across all datasets and coherence levels. These results demonstrate that coherence estimation window size selection is critically important in In-DMU applications: undersized windows are prone to noise contamination, whereas oversized windows obscure spatially heterogeneous features, thus undermining model estimation reliability.

Fig. 14(a) and (b) shows the density maps of deformation and coherence for the PDS and LD areas, respectively. As shown in Fig. 14, the deformation values are primarily concentrated around 0. Specifically, in the PDS area, the proportions of points falling within the $\pm 1\sigma$, $\pm 2\sigma$, and $\pm 3\sigma$ In-DMU estimated error ranges are 65.72%, 85.52%, and 92.01%, respectively, while in the LD area, the corresponding percentages are 74.67%, 96.41%, and 99.31%. Moreover, a marked dispersion in point density is observed when coherence drops below 0.65. These results validate the effectiveness of the constructed model.

VI. Conclusion

The precise quantification of InSAR deformation monitoring uncertainty serves as a fundamental prerequisite to ensure reliability in practical applications. In this study, a multiparameter-controlled experimental framework was developed to systematically analyze the combined effects of key factors—wavelength, coherence, spatial resolution, and deformation magnitude—on the precision of deformation inversion. Building upon these foundations, the universal

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wavelength–coherence deformation uncertainty model (In-DMU) was proposed for a priori precision estimation. Furthermore, the applicability boundary of the In-DMU model was defined, i.e., the spatial deformation gradient falls below the critical threshold of 6.433 mm/m for the L band, 1.407 mm/m for the C band, and 0.456 mm/m for the X band. Finally, validation using multisource SAR datasets (ALOS-1, TerraSAR-X, Radarsat-2, Sentinel-1, and ALOS-2) confirmed that the model performs well under diverse conditions.

The proposed In-DMU model requires only two input parameters for practical applications: wavelength and coherence level. The wavelength, as an inherent SAR satellite parameter, is readily available, while coherence can be estimated based on the relationship between surface characteristics (e.g., NDVI) and signal correlation. Although deformation monitoring precision remains unaffected by spatial resolution within the model's applicable range, high-resolution SAR images can reduce interferometric phase noise through multilooking processing, thereby improving practical monitoring precision.

The In-DMU model enables estimation of the precision of the deformation measurements before SAR data processing. However, further improvements are required. For example, a systematic evaluation of phase filtering algorithms and unwrapping methodologies on model performance as well as the development of temporal precision prediction frameworks for time-series InSAR applications. These developments would improve its applicability and reliability.

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